|| Jai Sri Gurudev ||

ADICHUNCHANAGIRI UNIVERSITY
FACULTY OF ENGINEERING, MANAGEMENT & TECHNOLOGY
(BGS INSTITUTE OF TECHNOLOGY)

BG Nagara – 571448, Nagamangala Taluk,

Mandya District, Karnataka (INDIA).

A Technical report
on

"SMART BLIND STICK WITH ADAPTIVE MULTIMODAL SENSING AND CONDITIONAL AI VISION ASSISTANCE"

Bachelor of Engineering

In

ELECTRONICS AND COMMUNICATION ENGINEERING

Submitted by:

NAZIF GANGAWALI	**22ECE043**
HARSHA K G	**22ECE048**
BHUVAN M S	**22ECE012**
HAMSIKA M K	**22ECE047**
KANNIKA V	**22ECE062**
CHANDANA J K	**22ECE018**

Under the Mentoring of:

Dr. NAVEEN K B , B.E, M.Tech (Ph.D)
HOD, Dept. of ECE,
BGSIT, BG Nagara, Mandya.

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

FACULTY OF ENGINEERING, MANAGEMENT & TECHNOLOGY
(BGS INSTITUTE OF TECHNOLOGY)

BG NAGARA – 571448

2025-2026

ACKNOWLEDGEMENT

At first, we would like to thank the Divine Soul **PADMABHUSHAN SRI SRI SRI DR. BALAGANGADARANATHA MAHASWAMIJI,** Founder President, Sri Adichunchanagiri Math and **SRI SRI SRI DR. NIRMALANANDANATHA MAHASWAMIJI**, President, Adichunchanagiri Shikshana Trust, for providing us with such excellent facilities and Lectures, without which, this hackathon work could not have acquired the shape it has now done.

We would like sincere gratitude to our beloved principal **DR. B N SHOBHA,** Dean & Principal, BGSIT, B.G Nagara for providing the necessary facilities and support, thus making it possible for us to obtain the necessary resources required to complete this hackathon work.

We wish to express my gratitude to **DR. M B ANANDARAJU,** Professor and Director of HRD**,** ACU**,** B G Nagara for helping to carry out this hackathon work.

We would like sincerely thank to **DR. NAVEEN K B,** Professor and Head of the Department, Electronics and Communication Engineering BGSIT, BG Nagar, for their technical guidance , cheerful encouragement support and valuable suggestion to complete this work.

Lastly, we take this opportunity to extend my full-hearted thanks, gratitude, and respect to our **Parents, Lecturers,** and all my **friends**, for giving me valuable advice and support at all times in all possible ways.

NAZIF GANGAWALI	**22ECE043**
HARSHA K G	**22ECE048**
BHUVAN M S	**22ECE012**
HAMSIKA M K	**22ECE047**
KANNIKA V	**22ECE062**
CHANDANA J K	**22ECE018**

ABSTRACT

This report presents a smart blind stick prototype designed to improve mobility, safety, and situational awareness for visually impaired users. The system combines dual-level time-of-flight obstacle sensing, inertial orientation estimation, GPS-based location tracking, multi-path alert communication, and event-driven vision assistance. On the stick side, an ESP32 works with a Nano-based trigger stage, foot-level and head-level distance sensing, GPS positioning, GSM alerts, and an NRF24L01 fallback communication link. On the goggles side, an ESP32-CAM is activated only when the sensing system detects a relevant event, thereby reducing power use and avoiding continuous camera operation. Captured images are sent to a phone-based application for AI processing through a primary cloud-based recognition path and a fallback ONNX model when the cloud path is unavailable. The proposed architecture aims to provide a low-power, adaptive, and reliable assistive platform for real-time obstacle warning, identity/object recognition, and emergency alerting.

Keywords:

Smart blind stick, ESP32, ESP32-CAM, ToF sensor, GPS, GSM, NRF24L01, ONNX, assistive mobility, adaptive vision.

INTRODUCTION

The proposed system is an assistive mobility solution for visually impaired users. It combines proximity sensing, inertial awareness, location tracking, emergency communication, and on-demand vision processing into a single platform. The design goal is not only to detect obstacles, but also to reduce false usage of power-hungry modules, provide fallback communication when a primary channel fails, and support image-based understanding only when required.



Problem Statement

Conventional blind sticks are usually limited to basic obstacle detection and do not always provide layered warning, reliable communication, or intelligent vision assistance. In many situations, a user needs more than a simple vibration or beep: the system should detect obstacles at different heights, share location information, escalate emergencies, and assist with recognition of surrounding scenes. At the same time, continuously running a camera and AI pipeline can increase power consumption and reduce battery life. The project therefore addresses the need for an adaptive, low-power, and fail-safe assistive system with multiple sensing and communication layers.

Objectives

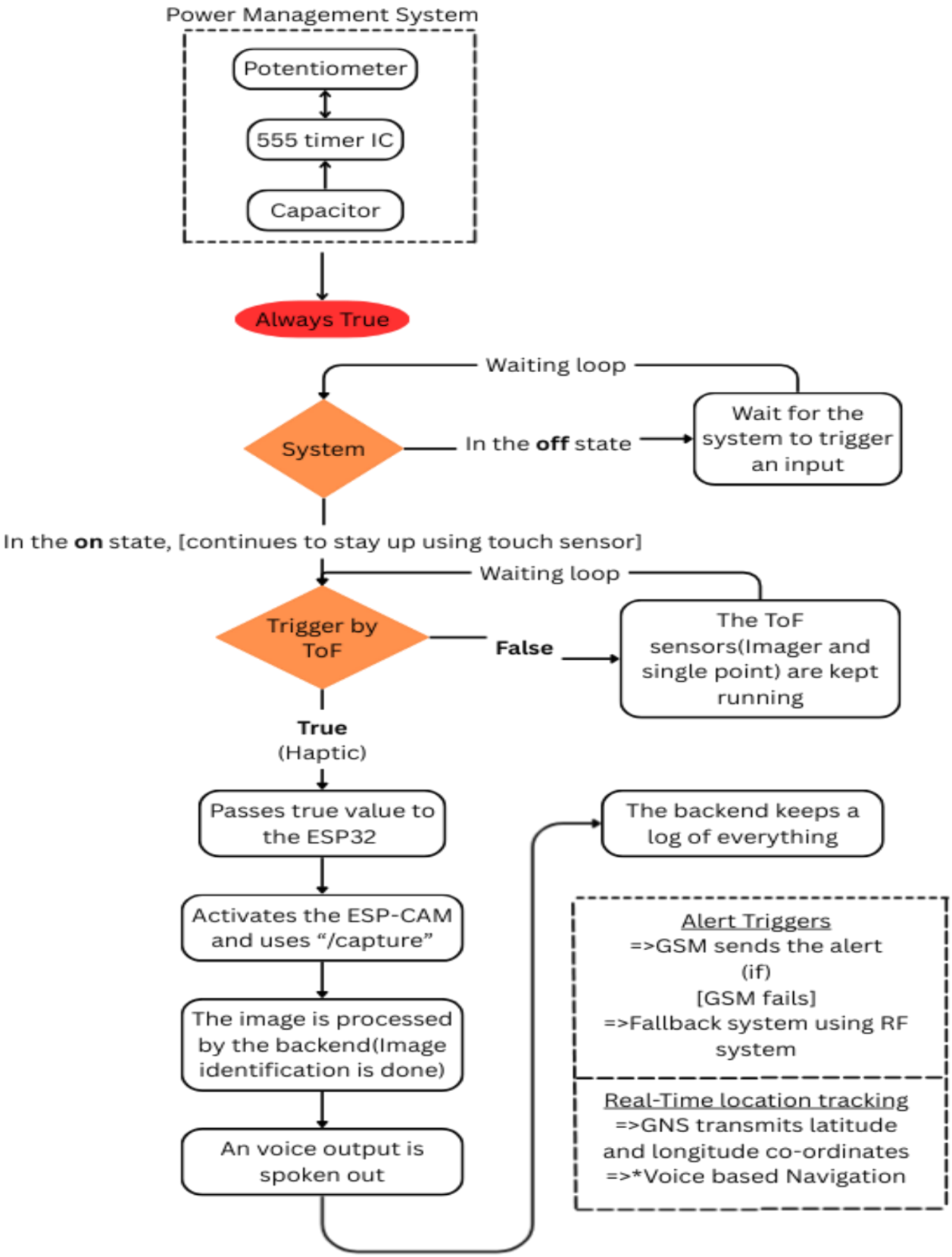
- Detect obstacles at foot level and head level using separate sensing logic.
- Estimate stick orientation using IMU-based angle monitoring to improve handling and calibration.
- Provide GPS-based location capture with fallback through mobile location services when hardware GPS is unavailable.
- Transmit alerts through GSM, with NRF24L01 as a local RF fallback path.
- Use touch-based inactivity detection to shut the stick down after a safe delay and save power.
- Activate the ESP32-CAM only when obstacle-related sensing requires visual confirmation.
- Send the captured image to a phone application for AI recognition through a primary cloud path and a local ONNX fallback.
- Deliver spoken output in real time to the user.

System Overview

The architecture is divided into two cooperative subsystems: the blind stick module and the goggles/vision module. The blind stick module contains the main sensing and communication hardware, while the goggles module contains the camera and the visual recognition pipeline. A mobile phone acts as the communication bridge and AI interface. When both ESP-based devices connect to the same hotspot, local communication occurs over the local network, while the phone uses internet data only for cloud-based recognition services.

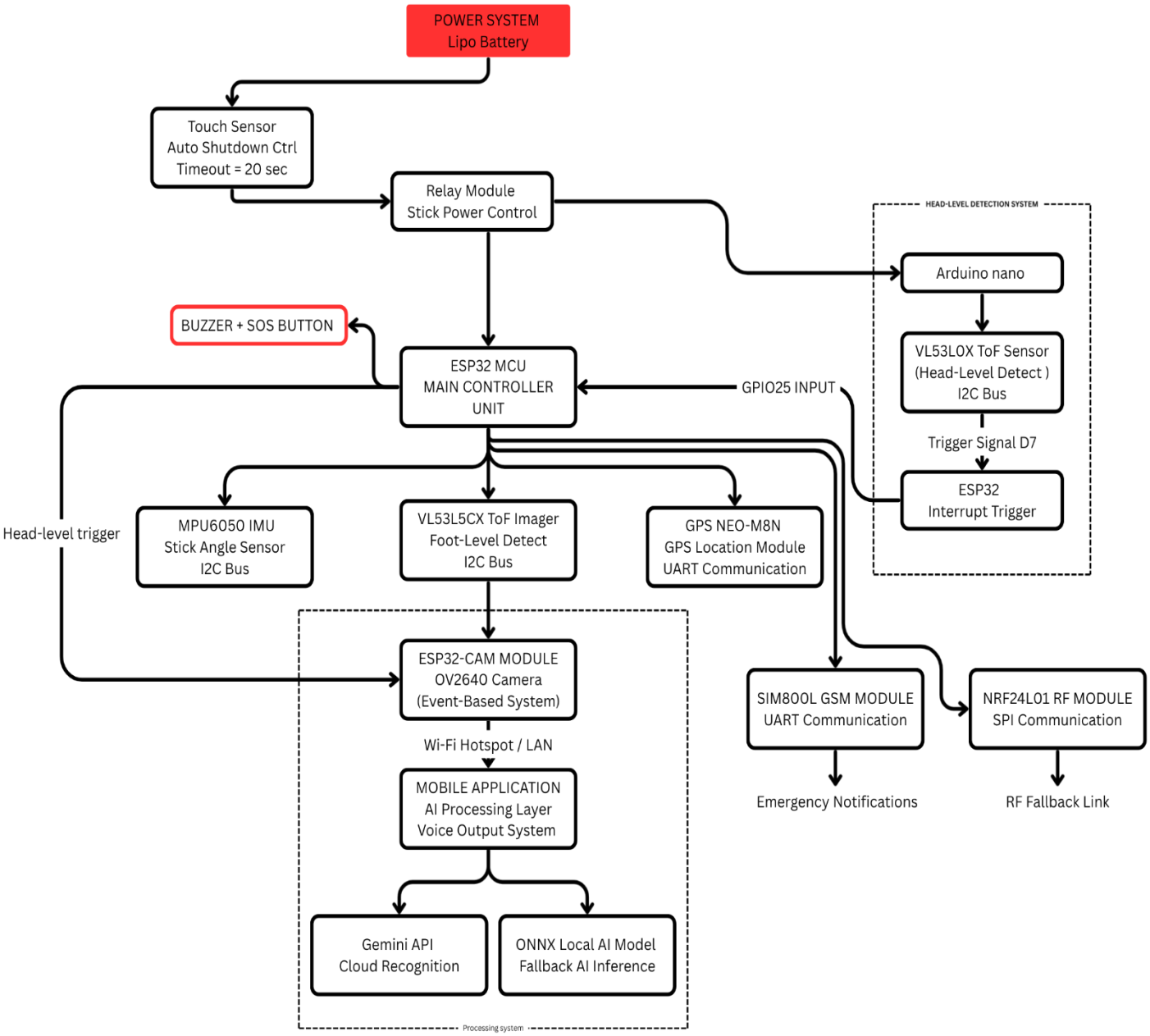
Hardware Architecture

Module	Role
ESP32 Dev Module	Main controller for sensors, alerts, and system coordination
Arduino Nano	Handles VL53L0X single-point sensing and sends trigger to ESP32
VL53L5CX ToF Imager	Detects foot-level obstacles and nearby scene depth
VL53L0X ToF Single Point	Detects head-level obstacles
MPU6050	Provides stick tilt information in X and Y axes
NEO-M8N GPS	Provides latitude and longitude from on-board GPS
SIM800L GSM	Sends SMS/call-based alert notifications
NRF24L01	Fallback RF communication link
Touch Sensor	Detects whether the stick is being held and enables auto shutdown logic
ESP32-CAM	Captures images for event-driven visual recognition
Mobile App	Runs AI inference routing and spoken output


Power Management System
Potentiometer
555 timer IC
Capacitor
Always True
Waiting loop
System
In the off state
Wait for the system to trigger an input
In the on state, [continues to stay up using touch sensor]
Waiting loop
Trigger by ToF
False
The ToF sensors(Imager and single point) are kept running
True
(Haptic)
Passes true value to the ESP32
Activates the ESP-CAM and uses "/capture"
The image is processed by the backend(Image identification is done)
An voice output is spoken out
The backend keeps a log of everything
Alert Triggers
=>GSM sends the alert
(if)
[GSM fails]
=>Fallback system using RF system
Real-Time location tracking
=>GNS transmits latitude and longitude co-ordinates
=>*Voice based Navigation

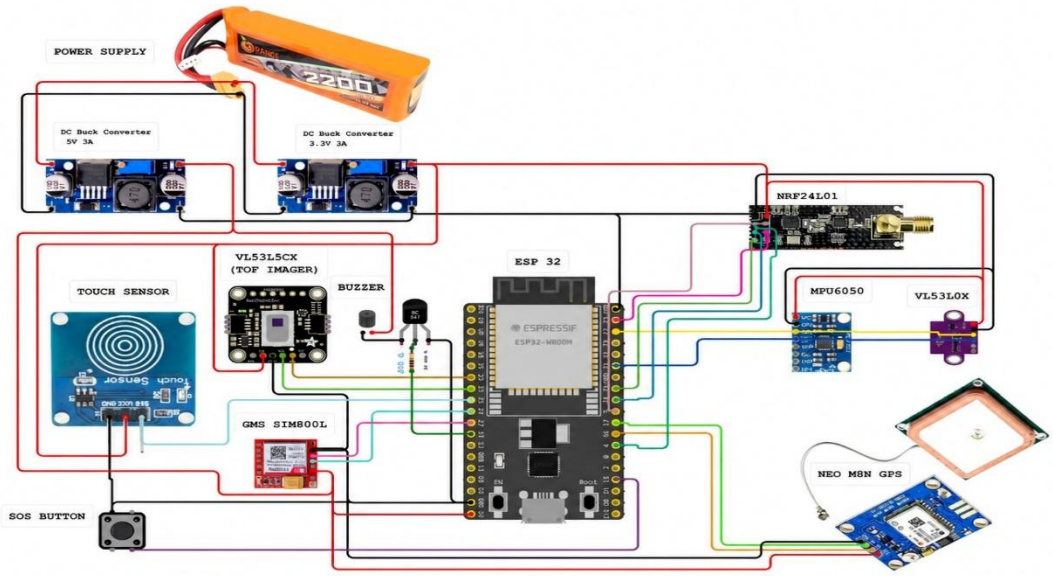
Communication Architecture

The design uses multiple communication paths for reliability. The primary path for user alerts is GSM. If GSM is unavailable or weak, the NRF24L01 link provides a local fallback channel. The stick and goggles can communicate through the same phone hotspot network, allowing low-latency local data exchange without using mobile data for device-to-device traffic. The phone itself can still use mobile data for cloud AI APIs.



Working Principle

1. The system powers on and initializes the ESP32, Nano, sensors, and communication modules.
2. The touch sensor monitors usage. If the stick is not held for the configured delay, the shutdown routine is enabled.
3. The VL53L5CX detects the nearby ground and foot-level hazards, while the VL53L0X monitors head-level hazards.
4. The Nano can generate a trigger signal to the ESP32 when its sensing logic detects an event.
5. The MPU6050 estimates the tilt angle so that the software can judge whether the stick is being held near the intended orientation.
6. When an obstacle event is confirmed, the ESP32-CAM is enabled to capture a frame.
7. The captured image is passed to the mobile application for AI recognition using the primary cloud model.
8. If the cloud path fails, the local ONNX model is used as a fallback.
9. The output is spoken to the user in real time, while alerts can also be transmitted through GSM or RF fallback.



Software Logic

The software is organized around event-driven states rather than continuous full-load operation. The idle state keeps only the necessary sensing and supervision active. When a sensor threshold is crossed, the system enters a warning state and then activates the vision state if visual confirmation is required. The recognition state first tries the cloud route, then switches to the local ONNX model if the primary route fails. The alert state sends messages through GSM or NRF24L01 depending on network availability.

Power Management

To conserve battery, the system uses a touch-based hold detection method. If the stick remains unattended for the configured period, the controller transitions into a shutdown state. The camera is not kept on continuously; instead, it is activated only when sensing indicates a relevant obstacle or scene event. This event-driven operation reduces power draw and extends usable battery life.

Prototype Implementation Notes

- The stick and goggles are treated as separate modules that cooperate through the phone network.
- The design supports both local communication and cloud-assisted recognition.
- The Nano-based trigger path isolates the single-point ToF sensing from the main ESP32 logic.
- The fallback system is intended to maintain usability when a module, network, or cloud service is temporarily unavailable.

Advantages

- Dual-level sensing improves awareness of both ground and upper-body obstacles.
- Conditional camera activation reduces battery consumption.
- Multiple communication paths improve reliability.
- Cloud-to-local AI failover improves continuity of service.
- Touch-based shutdown protects against unnecessary power drain.
- The system supports spoken feedback, making it user-friendly.

Comparison with Existing Systems

Existing assistive mobility devices for visually impaired individuals range from conventional white canes to electronically enhanced smart sticks. Traditional white canes rely solely on tactile feedback and are limited to detecting obstacles only at ground level. To overcome these limitations, several smart stick solutions using ultrasonic sensors have been proposed, which provide distance-based alerts but still lack environmental context and multi-level sensing.

Recent advancements have introduced camera-based assistive systems capable of object recognition using artificial intelligence. However, these systems often rely on continuous camera operation, resulting in high power consumption and dependency on stable internet connectivity. Additionally, many existing systems lack communication redundancy and fail to operate effectively under network failure conditions.

Comparative Analysis:

Feature	Traditional Stick	Ultrasonic Smart Stick	Camera-Based Systems	Proposed System
Foot-level detection	Yes	Yes	Yes	Yes
Head-level detection	No	No	Yes	Yes

Multi-sensor fusion	No	No	Limited	Yes
AI-based recognition	No	No	Yes	Yes
Conditional camera usage	No	No	No	Yes
Offline AI fallback	No	No	No	Yes
Communication fallback	No	No	No	Yes
Power Optimization	No	Limited	No	Yes

The proposed system introduces several key improvements over existing solutions. Unlike conventional systems, it incorporates multi-level sensing using both time-of-flight sensors for foot-level and head-level obstacle detection. The system employs a conditional vision activation mechanism, where the camera is triggered only when necessary, significantly reducing power consumption.

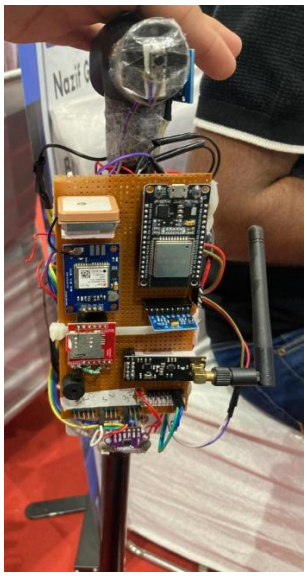
Furthermore, the integration of a hybrid AI pipeline ensures reliability by switching between a cloud-based recognition service and a locally stored ONNX model in case of connectivity issues. The inclusion of dual communication pathways, comprising GSM and RF modules, enhances system robustness in emergency situations. These features collectively make the proposed system more efficient, reliable, and practical for real-world deployment.

The proposed system is an assistive mobility solution for visually impaired users. It combines proximity sensing, inertial awareness, location tracking, emergency communication, and on-demand vision processing into a single platform. The design goal is not only to detect obstacles, but also to reduce false usage of power-hungry modules, provide fallback communication when a primary channel fails, and support image-based understanding only when required.

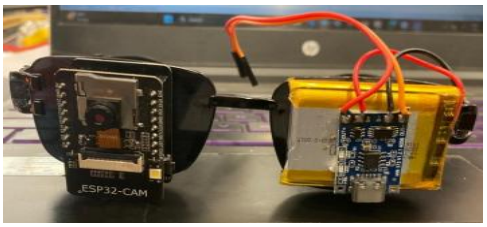
CONCLUSION

The proposed smart blind stick is an adaptive assistive system that combines layered sensing, conditional vision, redundant communication, and AI-based recognition. The architecture is suitable for a technical report submission and, after proper claim refinement and novelty review, can also support a patent application.

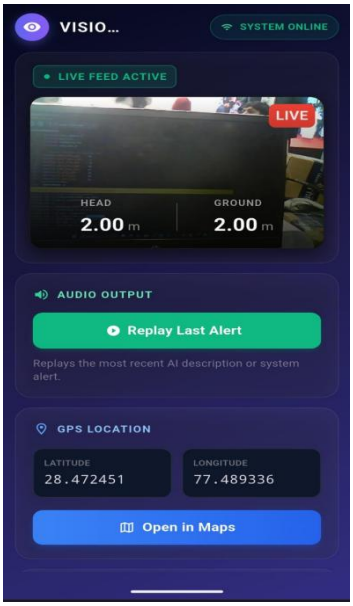
Results



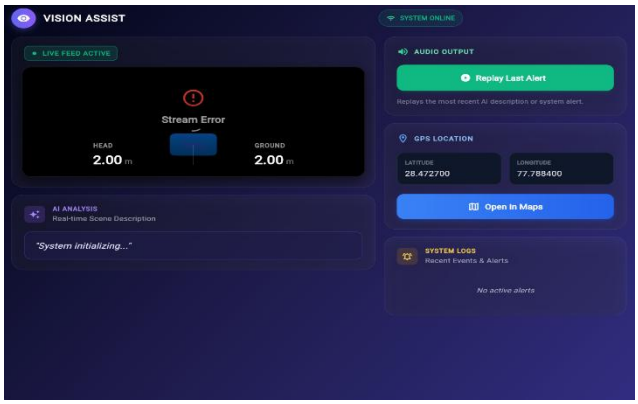
Smart Stick (Prototype)



Smart Goggles (Prototype)



Mobile Application



Web Application Dashboard